SUPRIYA SARKER

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EDUCATION

University of Arkansas, Fayetteville — *January 2025 - May 2028 (Expected)*
Ph.D. in Industrial Engineering

University of Tennessee, Knoxville — *August 2023 - December 2024*
MS in Computer Engineering

Chittagong University of Engineering & Technology — *June 2016 - February 2022*
MS in Computer Science & Engineering

Chittagong University of Engineering & Technology — *March 2010 - December 2014*
BS in Computer Science & Engineering

HONORS AND AWARDS

P.E.O. Scholar Award Nominee — *P.E.O. Sisterhood, 2027–2028*
P.K. Kuroda Endowed Graduate Fellowship in Engineering — *University of Arkansas, 2026–2027*
Best Paper Award – Honorable Mention — *ICAICT, 2020*
Undergraduate Academic Merit Scholarship, CUET — *March 2010 - November 2014*

SOFTWARE AND TOOLS

Programming Languages and Tools	Python, SQL, C, Java, MATLAB, HTML, LaTeX
Development Tools	Git/GitHub, Jupyter Notebook, Google Colab, Excel
AI/LLM Frameworks	PyTorch, TensorFlow, Hugging Face, LangChain, Vector Databases, Prompt Engineering, LLM API Integration
Data Science Libraries	Pandas, NumPy, SciPy, Matplotlib, Scikit-learn

RESEARCH METHODOLOGIES

Machine Learning, Deep Learning, Reinforcement Learning, Generative AI, Multimodal AI, Agentic AI Systems, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

RESEARCH EXPERIENCES

University of Arkansas, Fayetteville — August 2026 - Present
Graduate Research Assistant

- Developing a context-aware LLM framework using RAG and plant-specific risk data to prioritize maintenance work orders in nuclear power plants.

University of Tennessee, Knoxville — January 2026 - July 2026
Graduate Research Assistant

- Designed an agentic AI system leveraging large language models to support autonomous inspection and quality assessment in manufacturing, integrating multi-step reasoning with multimodal sensor data and domain-specific rules to detect, interpret, and classify metal deposition anomalies.

Oak Ridge National Laboratory — August 2025 - December 2025
Graduate Research Intern

- Designed a heterogeneous data-aware, privacy-preserving hyperparameter optimization framework for federated learning with large foundational models by integrating adaptive trial selection and privacy budget allocation, enabling controlled trade-offs between model performance, communication overhead, and cumulative differential privacy loss.
- Conducted empirical evaluation on Vision Transformer (ViT) models using MNIST, analyzing the impact of hyperparameters on model performance under differential privacy constraints.

University of Tennessee, Knoxville — January 2025 - July 2025
Graduate Research Assistant

- Designed a transparent decision support framework that integrates predictive modeling and LLM-generated patient-friendly explanations to enhance trust and transparency in clinical decision-making.

University of Tennessee, Knoxville August 2023 - December 2024
Graduate Teaching and Research Assistant

- Built the Beacon real-world blackout traffic dataset and applied Rainbow DQN to control robot vehicles in reconstructed traffic, reducing wait times by up to 82.6%. **GitHub**

Chittagong University of Engineering & Technology March 2017 - February 2022
Graduate Research Assistant

- Built an LSTM classifier for seven driving maneuvers using sensor dataset, achieving over 98% per-class accuracy. **GitHub**

Chittagong University of Engineering & Technology March 2014 - December 2014
Graduate Research Assistant

- Developed a hybrid digital watermarking algorithm using Discrete Wavelet Transform to enhance copyright protection, focusing on robustness against noise and compression while preserving image quality.

PROFESSIONAL EXPERIENCE

World University of Bangladesh, Dhaka, Bangladesh May 2018 - August 2022
Lecturer, Department of Computer Science & Engineering

- Taught core undergraduate computer science courses, supervised student projects and theses, and conducted research leading to peer-reviewed publications.

Comilla Model College, Comilla, Bangladesh December 2015 - September 2016
Lecturer, Department of Information & Communication Technology

- Delivered fundamental ICT coursework, designed and graded assessments to evaluate student learning outcomes.

Multiway Solutions LTD, Dhaka, Bangladesh January 2015 - November 2015
Software Engineer

- Designed and implemented web-based applications for business clients using PHP and MySQL to streamline operations and meet client-specific requirements.

SELECTED CONFERENCE PUBLICATIONS

Supriya Sarker, Iftekharul Islam, Bibek Poudel, Weizi Li. Beacon: A Naturalistic Driving Dataset During Blackouts for Traffic Reconstruction and Control Benchmarking. *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.

Supriya Sarker, and Md Mokammel Haque. A Data Change Rule Based Empirical Framework for Labeling Unlabeled Time Series Driving Data. In: *2020 2nd International Conference on Advanced Information and Communication Technology (ICAICT)*, IEEE, 2020, pp. 476-479.

Supriya Sarker, and Md Mokammel Haque. An Approach Towards Domain Knowledge-Based Classification of Driving Maneuvers with LSTM Network. In: M.S. Uddin, J.C. Bansal (eds) Proceedings of International Joint Conference on Advances in Computational Intelligence. *Algorithms for Intelligent Systems*, Springer, Singapore, 2021.

Supriya Sarker, Md Sajedur Rahman, Ashraful Alam, Md Jahirul Islam, Dipta Sikder. Energy saving smart waste segregation and notification system. In: *2020 IEEE Region 10 Symposium (TENSYMP)*, IEEE, 2020, pp. 275-278.

Supriya Sarker, Md Sajedur Rahman, Md. Nazmus Sakib. An approach towards intelligent accident detection, location tracking and notification system. In: *2019 IEEE international conference on telecommunications and photonics (ICTP)*, IEEE, 2019, pp. 1-4.

Supriya Sarker, Md. Shahraduan Mazumder, Md Sajedur Rahman, Md Anayt Rabbi. An assistive HCI system based on block scanning objects using eye blinks. In: *2019 4th International Conference on Electrical Information and Communication Technology (EICT)*, pp. 1-6.

SELECTED JOURNAL PUBLICATIONS

Supriya Sarker, Dr. Mohammad Shamsul Arefin, Md Kowsher, Touhid Bhuiyan, Pranab Kumar Dhar, Oh-Jin Kwon, A Comprehensive Review on Big Data for Industries: Challenges and Opportunities, *IEEE Access*. 2022

Md Mokammel Haque, **Supriya Sarker**, M Ali Akber Dewan. Driving Maneuver Classification From Time Series Data: A Rule Based Machine Learning Approach. *Applied Intelligence*, pp. 1-16, 2022.

Supriya Sarker, Md Mokammel Haque, M Ali Akber Dewan. Driving Maneuver Classification Using Domain Specific Knowledge and Transfer Learning. *IEEE Access*, vol. 9, pp. 86590 – 86606, June, 2021.

PUBLICATIONS UNDER REVIEW

Supriya Sarker, Beau Schelble, Andrew Farrell, Anuj Kapadia, Westyn Branch-Elliman, Ritesh Ramdhani, Anahita Khojandi. Reimagining the Role of Human-AI Teaming in Healthcare: A Framework for the Future of Outpatient Care. *AI and Ethics*. October 2026.